

Supplementary Technical Documentation

1. Overview

This toolbox provides a unified framework for unbiased estimation of Maximum Mean Discrepancy (MMD) variance. It addresses key challenges in kernel-based hypothesis testing:

Unbiasedness: Reliable performance under both the Null Hypothesis and Alternative Hypothesis.

Imbalanced Data: Robust performance even when sample sizes are unequal $n \neq m$.

Exact Acceleration: Reduces complexity from $O(n^2)$ to $O(n \log n)$ for univariate Laplacian kernels.

2. Core algorithms

‘MMD_propose.m’: The proposed unbiased variance estimator.

‘euMMD.m’: The exact acceleration engine. It utilizes linear prefix sums and sorting to achieve quasi-linear time complexity without any numerical approximation.

3. Numerical experiments

Reproduce Section 4 of the paper:

1. Variance analysis under heterogeneous settings (Section 4.1)

‘sec1_caseA_null_balanced.m’: the Null hypothesis, balanced samples (Table 1).

‘sec1_caseB_alt_balanced.m’: the Alternative hypothesis, balanced samples (Table 2).

‘sec1_caseC_null_unbalanced.m’: the Null hypothesis, unbalanced samples (Table 3).

‘sec1_caseD_alt_unbalanced.m’: the Alternative hypothesis, unbalanced samples (Table 4).

2. Properties of the estimator (Section 4.2)

‘sec2_property1.m’: verifies the exact algebraic equivalence of the statistics (Figure 2).

‘sec2_property2.m’: tracks the empirical variance decay profiles (Figure 3).

3. Accuracy and efficiency of Laplacian-based accelerated euMMD (Section 4.3)

‘sec3_compare_accuracy.m’: compares MMD and euMMD (Table 5).

‘sec3_step1_benchmark_time_efficiency.m’: measures execution time scaling.

‘sec3_step2_benchmark_space_eummd.m’: measures peak memory for $O(n^2)$ matrix-form.

‘sec3_step3_benchmark_space_matrix.m’: measures peak memory for the $O(n \log n)$ euMMD.

‘sec3_step4_plot_efficiency.m’: (Figure 4).

4. Baselines for comparison

This toolbox includes implementations of state-of-the-art MMD variance estimators:

‘MMD_gretton.m’: [\[Gretton et al. \(2012\)\]](#).

‘MMD_sutherland.m’: [\[Sutherland & Deka \(2019\)\]](#).

‘MMD_wei.m’: [\[Wei et al. \(2025\)\]](#).